

DRONE ROOF CLEANING SOP

FLYBY SHINE

1. PURPOSE

This document establishes the operational framework, chemical education standards, and liability awareness principles for drone-assisted roof cleaning performed by Flyby Shine.

The intent of this SOP is to provide educational guidance on commonly accepted industry practices while clearly defining risk boundaries and responsibility. This document does not replace manufacturer instructions, Safety Data Sheets (SDS), regulatory requirements, or job-specific execution plans.

2. SCOPE

This SOP applies exclusively to drone-assisted roof cleaning on commercial and institutional roof systems, including asphalt shingle, metal, tile, membrane, and coated roofs.

This SOP does not apply to ground-based cleaning methods, manual roof access, or residential walkable roof systems unless explicitly authorized by a separate document.

3. AUTHORITY AND LIMITATIONS

This SOP provides operational guidance and industry education only. It does not:

- Authorize regulatory compliance claims
- Guarantee outcomes or performance
- Override manufacturer instructions or warranties
- Approve undocumented or unsafe application practices

Aircraft limitations, chemical manufacturer guidance, SDS documentation, and property-specific requirements always take precedence over this SOP.

4. RISK AND LIABILITY AWARENESS

Drone roof cleaning introduces inherent risk related to roof height, pitch, condition, public proximity, weather variability, environmental exposure, and chemical selection.

As operational risk increases, conservative decision-making is required. Liability exposure increases when chemical application exceeds manufacturer guidance, is applied without regard to roof condition, or fails to account for environmental sensitivity.

This SOP is designed to inform decision-making, not remove responsibility from the executing party.

5. CHEMICAL SELECTION AND RATIO EDUCATION (INDUSTRY STANDARDS)

Roof cleaning chemistry is not universal. However, the professional roof cleaning industry recognizes common chemical categories and concentration ranges used as starting points by trained operators.

All concentration ranges referenced in this SOP are provided for educational purposes only. Final application variables must be determined by the operator based on manufacturer documentation, roof condition, environmental factors, and site-specific risk.

6. ROOF-TYPE CHEMICAL STANDARDS

ASPHALT SHINGLE ROOFS

Common Chemical Category:

Sodium hypochlorite-based solutions with appropriate surfactants.

Typical Industry Concentration Range (On-Roof):

Minimum effective concentration commonly begins at approximately 4 percent sodium hypochlorite.

Standard working range commonly falls between 4 percent and 6 percent sodium hypochlorite.

Severe organic growth conditions may require concentrations exceeding 6 percent, subject to roof condition and operator judgment.

Primary Liability Considerations:

Granule loss

Accelerated shingle degradation

Potential manufacturer warranty voidance

Operator Considerations:

Older, brittle, or previously treated shingles may require more conservative approaches.

Increasing concentration increases cleaning effectiveness while simultaneously increasing long-term damage risk.

METAL ROOF SYSTEMS

Common Chemical Category:

Mild oxidizers or manufacturer-approved cleaners.

Typical Industry Concentration Range:

Approximately 2 percent to 4 percent sodium hypochlorite when permitted by manufacturer guidance and coating specifications.

Primary Liability Considerations:

Oxidation

Finish degradation

Galvanic reactions

Operator Considerations:

Painted, coated, and bare metal systems respond differently to chemical exposure.

Manufacturer approval is critical prior to application.

TILE ROOF SYSTEMS

Common Chemical Category:

Neutral to mildly alkaline cleaners or low-to-moderate oxidizers.

Typical Industry Concentration Range:

Approximately 3 percent to 5 percent sodium hypochlorite, adjusted conservatively based on tile condition and porosity.

Primary Liability Considerations:

Surface etching

Discoloration

Tile breakage

Operator Considerations:

Tile age, porosity, and prior chemical exposure materially affect outcomes.

MEMBRANE AND COATED ROOF SYSTEMS

Common Chemical Category:

Manufacturer-approved cleaners only.

Typical Industry Concentration Range:

Defined exclusively by manufacturer documentation.

Primary Liability Considerations:

Coating failure

Seam degradation

Warranty avoidance

Operator Considerations:

If manufacturer guidance is unavailable, restrictive, or unclear, cleaning must be refused.

7. WHY RATIOS VARY AND WHERE LIABILITY RESIDES

Chemical ratios vary due to roof material, roof condition, severity of organic growth, ambient temperature, dwell exposure, runoff sensitivity, and manufacturer warranty constraints.

Improper chemical selection or dilution can result in permanent roof damage, environmental harm, denied warranty claims, and third-party liability.

For this reason, Flyby Shine treats all ratio information as educational reference only. Final responsibility for chemical selection and application resides with the executing party.

8. OPERATOR RESPONSIBILITY

Operators are responsible for reviewing manufacturer documentation, SDS guidance, and roof system specifications prior to application.

Operators must select products approved for the specific roof system, adjust application variables based on real-time site conditions, and document decisions in the job record.

This SOP does not authorize deviation from manufacturer requirements.

9. DRONE APPLICATION FRAMEWORK

Drone-assisted roof cleaning is conducted using low-impact application principles designed to reduce human exposure and minimize surface contact.

Key considerations include controlled application appropriate to roof geometry, avoidance of aggressive mechanical force, and continuous situational awareness during flight operations.

Specific application mechanics are governed by internal technical references and job-specific execution plans.

10. ENVIRONMENTAL AND PROPERTY PROTECTION

Operators must account for overspray drift, runoff pathways, vegetation exposure, and adjacent property protection.

Environmental exposure is treated as a primary risk factor, not a secondary concern.

11. STOP-WORK AND ABORT AUTHORITY

Any operator may pause or abort operations if conditions change or risk cannot be adequately controlled.

Examples include weather instability, equipment irregularities, unexpected public exposure, or loss of situational awareness.

When uncertainty exists, operations must stop.

12. VERIFICATION AND DOCUMENTATION

A project is considered complete when organic growth has been visibly reduced or neutralized, no apparent surface damage is observed, surrounding areas are left in acceptable condition, and pre- and post-operation imagery is captured.

Documentation supports transparency and auditability and does not constitute a performance guarantee.

13. REFUSAL CONDITIONS

Operations must be refused or halted if requested methods conflict with manufacturer guidance, chemical use exceeds documented approvals without justification, environmental or public risk cannot be mitigated, or instructions fall outside approved execution plans.

14. DISCLAIMER

This SOP is provided for educational and operational guidance only. It does not constitute legal advice, regulatory approval, safety certification, or manufacturer endorsement.

Responsibility for compliance with all applicable laws, manufacturer instructions, and safety standards rests with the executing party.